

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5 TOTAL MARKS: 50

Annexure A

Comparative Analysis

Code of Conduct:

I p.paneendra (192321072) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: p.paneendra

Annexure A
Comparative Analysis

1. Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.
2. Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.
3. Perform a comparative analysis of Pretty Good Privacy (PGP) and S/MIME in terms of key management, trust models, scalability, and enterprise usability. Recommend the most suitable approach for a corporate email system.
4. Compare rule-based and machine learning-based email spam detection techniques. Analyze their effectiveness in handling evolving spam patterns and minimizing false positives.
5. Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.

1. Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec

Authentication Header (AH) and Encapsulating Security Payload (ESP) are security protocols used in IPSec to protect data transmitted over an IP network. AH mainly provides data integrity, data authentication, and protection against replay attacks. It ensures that the packet has not been modified during transmission and confirms the authenticity of the sender. However, AH does not provide confidentiality because it does not encrypt the actual data. Therefore, the contents of the communication can still be viewed by an attacker. ESP provides more comprehensive security services. It provides confidentiality through encryption, along with data integrity, authentication, and anti-replay protection. Because encryption requires additional processing, ESP generally has slightly higher overhead than AH. However, modern systems can perform encryption efficiently using hardware and optimized cryptographic algorithms. AH is useful when authentication and integrity are required but encryption is not necessary. ESP is more suitable when sensitive information needs to be protected from unauthorized access. For a VPN deployment, ESP is more appropriate because VPN communication usually travels through an untrusted public network such as the Internet. ESP can encrypt the communication and prevent attackers from reading sensitive information. It also supports authentication, integrity, and anti-replay protection. Conclusion: ESP is generally preferred for modern VPNs because it provides stronger and more complete security than AH.

2. Compare Transport Mode and Tunnel Mode in IPSec

IPSec provides two modes of operation: Transport Mode and Tunnel Mode. Both modes provide security for IP communication, but they are designed for different network scenarios. In Transport Mode, only the payload of the IP packet is protected. The original IP header remains visible because it is required for routing the packet between the source

and destination hosts. Transport mode has lower overhead because it does not encapsulate the entire original IP packet. Transport mode is mainly suitable for host-to-host communication. For example, two servers or two computers can use IPSec transport mode to communicate securely. It provides efficient protection when both communicating hosts directly support IPSec. In Tunnel Mode, the complete original IP packet, including its original IP header, is encapsulated inside a new IP packet. A new IP header is added for routing between the VPN endpoints. This provides stronger protection because the original IP addressing information is hidden from external networks. Tunnel mode is mainly used for network-to-network communication, especially in site-to-site VPNs. For example, an organization may have one private network in Chennai and another private network in Bangalore. VPN gateways can use IPSec tunnel mode to securely connect both networks over the Internet. Transport mode has less overhead and is efficient for direct host communication. Tunnel mode has additional overhead but provides better protection and is more suitable for VPN gateways. Conclusion: Transport Mode is generally suitable for host-to-host communication, while Tunnel Mode is more appropriate for network-to-network communication and VPN deployments.

3. Compare PGP and S/MIME Pretty Good Privacy (PGP) and S/MIME (Secure/Multipurpose Internet Mail Extensions) are security technologies used to protect email communication. Both provide encryption and digital signatures, but their approaches to key management and trust are different. PGP uses public-key cryptography to encrypt email messages and create digital signatures. It commonly uses a Web of Trust model. In this model, users can verify other users' public keys and digitally sign them to indicate that they trust those keys. The Web of Trust provides flexibility because it does not require a single central authority. However, it can become difficult to manage when there are thousands of users. Users need to understand how to create, exchange, verify,

and manage their keys. S/MIME also uses public-key cryptography but relies on digital certificates. These certificates are generally issued and verified through trusted Certificate Authorities (CAs). The certificate connects a user's identity with their public key. S/MIME is particularly suitable for organizations because certificates can be managed centrally. Corporate administrators can issue certificates to employees, renew them, revoke compromised certificates, and manage security policies. PGP provides greater flexibility and is suitable for individuals and decentralized environments. S/MIME provides better centralized management and integration with enterprise email systems. For a corporate email system, S/MIME is generally more suitable because large organizations require centralized certificate management, scalability, easier administration, and integration with existing enterprise email infrastructure. Conclusion: PGP is useful for decentralized and individual communication, while S/MIME is generally preferred for corporate email security.

4. Compare Rule-Based and Machine Learning-Based Email Spam Detection

Email spam detection is used to identify and block unwanted, fraudulent, or malicious emails. Two important approaches are rule-based detection and machine learning-based detection. In a rule-based system, predefined rules are created to identify spam emails. These rules may check the sender's address, suspicious keywords, number of links, email headers, blacklisted domains, or unusual message patterns. For example, a system may define a rule such as: if an email comes from a known blacklisted sender, classify it as spam. Similarly, emails containing certain suspicious words or large numbers of malicious links may be blocked. Rule-based detection is simple, fast, and easy to understand. However, it has a major limitation: it depends on manually created rules. If attackers change their spam techniques, existing rules may fail to identify the new spam. Machine learning-based spam detection uses algorithms that learn patterns from previously classified emails. Algorithms such as Naive Bayes,

Decision Trees, Random Forest, Support Vector Machines, and neural networks can be used. The system can analyze features such as email content, sender information, links, headers, attachments, and message structure. The trained model can then classify new emails as spam or legitimate. Machine learning is more adaptable than rule-based detection because it can learn new patterns from updated training data. However, it requires suitable datasets, training, monitoring, and periodic retraining. One important problem is the false positive. A false positive occurs when a legitimate email is incorrectly classified as spam. High-quality and balanced training data can help reduce this problem. For modern email systems, a hybrid approach combining rules and machine learning is often more effective. Rules can quickly identify known threats, while machine learning can detect new and changing spam patterns. Conclusion: Machine learning is more suitable for evolving spam because of its adaptability. A combination of rule-based and machine learning techniques can provide stronger and more reliable spam detection.

5. Compare Manual Key Management and Automated Key Management Using IKE in IPSec

Key management is an important part of IPSec because encryption keys are required to establish secure communication between devices. Keys can be managed manually or automatically using protocols such as Internet Key Exchange (IKE). In manual key management, the network administrator manually generates and configures security keys on the IPSec devices. Both communicating devices must have matching configuration information. Manual key management can be simple for a small network containing only a few devices. However, it becomes difficult when the number of VPN connections increases. Administrators must manually configure, update, replace, and protect keys on every device. Manual management also increases the possibility of human errors. If a key is compromised, administrators need to manually replace it on all affected devices. This makes large-scale key management time-consuming and difficult.

Automated key management using IKE solves many of these problems. IKE allows two IPSec devices to automatically negotiate security parameters and establish cryptographic keys. A simplified process is: Device A → IKE negotiation → Device B → Secure keys established → IPSec communication. IKE can automatically establish Security Associations and negotiate suitable cryptographic algorithms and keys. It can also support periodic key changes, which improves security. Automated key management is particularly important in large organizations where many routers, firewalls, servers, and VPN connections need to communicate securely. Although IKE requires some initial configuration and knowledge, it significantly reduces the long-term administrative workload. It also reduces configuration errors and makes key management easier to scale. Conclusion: Manual key management may be suitable for small networks and simple testing environments. For large organizations, automated key management using IKE is strongly recommended because it provides better scalability, improved security, automatic key negotiation, easier key rotation, and lower operational complexity.